Optimal strategies in the all-heads coin game

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Abstract

We study a sequential coin-flipping game in which a player starts with n coins, each landing heads independently with probability p. In each round the player flips all remaining coins and must set aside at least one coin showing heads; if no coin shows heads, the player loses. The player wins if and when all coins have been set aside. This is a stochastic shortest-path Markov decision process whose Bellman equation involves a nonlinear suffix-maximum operator. We analyse two natural strategies — One (set aside exactly one head) and All (set aside every head) — and determine the optimal winning probability $w_{n,p}$ as a function of n and p.

For $p = \frac{1}{2}$ every strategy that accepts the immediate win achieves $w_{n,1/2} = \frac{1}{2}$. For $p > \frac{1}{2}$ the strategy One is optimal and $n \mapsto w_{n,p}$ is strictly increasing. For $p < \frac{1}{2}$ near $\frac{1}{2}$ we carry out a first-order perturbation expansion in $\delta := \frac{1}{2} - p$ and prove that the leading coefficient c_n of the deficit $\frac{1}{2} - w_{n,p}$ satisfies a linear recursion for $n \geq 7$, with a limit $L = \lim c_n \approx 1.7035$. As a consequence the optimal value sequence has, to first order in δ, a strict local minimum at $n = 5$ and no local maximum — local maxima are a non-perturbative phenomenon.

1 Introduction

1.1 The game

Fix a probability $p \in (0, 1)$ and a positive integer n. A player is given n coins, each of which independently lands heads with probability p and tails with probability $q := 1-p$. The game proceeds in rounds. In each round:

(i) All remaining coins are flipped simultaneously.

(ii) If no coin shows heads, the game is lost.

(iii) Otherwise the player must set aside at least one coin that shows heads. The set-aside coins leave the game permanently.

(iv) If every coin has been set aside, the game is won. Otherwise, a new round begins with the remaining coins.

The player's objective is to maximise the probability of winning.

Two natural strategies present themselves:

- **Strategy One:** set aside exactly one head when $1 \le k \le n-1$ heads appear, and set aside all n when $k = n$.
- **Strategy All:** set aside every head that appears ($i = k$).

Strategy One is conservative: it preserves as many coins as possible for future rounds, betting that the one set-aside coin has already "done its job." Strategy All is greedy: it cashes in every success immediately. Neither is universally optimal.

A variant of this game, in which the objective is to maximise the expected number of heads collected rather than the probability of collecting all of them, was introduced and studied by van Doorn [7]. The all-heads-wins formulation studied here leads to a richer structural picture.

1.2 Markov decision processes and the Bellman equation

The game is a finite-state *Markov decision process* (MDP): the state is the number m of coins remaining, the action is the number $i \in \{1, \dots, k\}$ of heads to set aside (where k is the random number of heads), and the transition is $m \mapsto m - i$. There are two absorbing states: *win* ($m = 0$, all coins collected) and *lose* (reached when $k = 0$ in some round). A *policy* (or *strategy*) π specifies, for each state n and each observed number $k \in \{1, \dots, n\}$ of heads, the number $i_\pi(n, k) \in \{1, \dots, k\}$ of heads to set aside; we write v_n^π for the resulting winning probability starting from state n.

Because every policy absorbs with probability at least $p^m + (1-p)^m > 0$ per round, the MDP is a *stochastic shortest-path* (SSP) problem in the sense of Bertsekas and Tsitsiklis [2]: the Bellman operator is a contraction and the optimal value function is its unique fixed point.

Definition 1.1 (Optimal winning probability)**.** Let $w_{0,p} := 1$. For $n \ge 1$ define

$$w_{n,p} \;:=\; p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \max_{j \le m \le n-1} w_{m,p}, \tag{1}$$

where $q = 1 - p$ and the empty sum (at $n = 1$) gives $w_{1,p} = p$.

The term p^n accounts for the event that all n coins land heads (an immediate win). Each summand with $j \ge 1$ accounts for the event that exactly j coins land tails; the player then chooses to keep some number $m \in \{j, \dots, n-1\}$ of coins (setting aside $n - m$ heads), and the game continues from state m with optimal value $w_{m,p}$.

The Bellman equation (1) is the standard optimality equation for this SSP–MDP. For general background on Markov decision processes and the Bellman equation we refer to the monographs of Puterman [5], Bertsekas [1], and Stokey, Lucas and Prescott [6].

1.3 Strategies One and All

Under strategy One the player always keeps $m = n - 1$ coins (except when $k = n$), so the Bellman recursion collapses to

$$a_{n,p} \;=\; p^n + \big(1 - p^n - (1-p)^n\big)\, a_{n-1,p}, \qquad a_{0,p} := 1. \tag{2}$$

Under strategy ALL the player always keeps $m = j$ coins (the number of tails), giving

$$b_{n,p} \;=\; p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \, b_{j,p}, \qquad b_{0,p} := 1. \tag{3}$$

Note that the recursions for both a and b are *linear* (no max operator), which makes them analytically more tractable than the Bellman equation (1).

1.4 Outline

In Section 2 we prove that at $p = \frac{1}{2}$ the optimal winning probability is $w_{n,1/2} = \frac{1}{2}$ for every n, and that in fact every reasonable strategy achieves this value. In Section 3 we show that for $p > \frac{1}{2}$ the strategy ONE is always optimal and $n \mapsto w_{n,p}$ is strictly increasing. Section 4 treats the case $p < \frac{1}{2}$ near $\frac{1}{2}$: we introduce a first-order perturbation in $\delta = \frac{1}{2} - p$, derive a recursion for the leading coefficient c_n, prove that it simplifies to a linear recursion for $n \geq 7$, and compute the limit $L = \lim c_n$.

2 The fair coin: $p = \frac{1}{2}$

Theorem 2.1. *For every $n \geq 1$, $w_{n,1/2} = \frac{1}{2}$. More generally, for any policy π satisfying $i_\pi(n,n) = n$ for every n (i.e. the player accepts the immediate win when all coins land heads), the winning probability is $v_n^\pi = \frac{1}{2}$.*

In particular, both strategies ONE and ALL achieve the optimum: $a_{n,1/2} = b_{n,1/2} = w_{n,1/2} = \frac{1}{2}$.

Proof. We use the binomial identity

$$\sum_{j=1}^{n-1} \binom{n}{j} \left(\frac{1}{2}\right)^n \;=\; 1 - 2^{1-n}. \tag{4}$$

Step 1. We show $w_{n,1/2} = \frac{1}{2}$ by strong induction. For $n = 1$: $w_{1,1/2} = \frac{1}{2}$. For $n \geq 2$, suppose $w_{m,1/2} = \frac{1}{2}$ for all $1 \leq m \leq n-1$. Then every suffix-maximum equals $\frac{1}{2}$, so

$$\begin{aligned} w_{n,1/2} &= 2^{-n} + \frac{1}{2} \sum_{j=1}^{n-1} \binom{n}{j} 2^{-n} \\ &= 2^{-n} + \tfrac{1}{2}\big(1 - 2^{1-n}\big) \\ &= 2^{-n} + \tfrac{1}{2} - 2^{-n} \;=\; \tfrac{1}{2}. \end{aligned}$$

Step 2. For a general policy π with $i_\pi(n,n) = n$, conditioning on the number of heads $K \sim \mathrm{Bin}(n, \frac{1}{2})$ gives

$$v_n^\pi = 2^{-n} \cdot 1 + \sum_{k=1}^{n-1} \binom{n}{k} 2^{-n} \, v^\pi_{n - i_\pi(n,k)}.$$

Since $1 \leq i_\pi(n,k) \leq k \leq n-1$ for $k < n$, the subscript $n - i_\pi(n,k)$ lies in $\{1, \ldots, n-1\}$, and by strong induction each $v^\pi_{n-i_\pi(n,k)} = \frac{1}{2}$. The same calculation as in Step 1 gives $v_n^\pi = \frac{1}{2}$. $\square$

Remark 2.2. A policy that refuses the immediate win at $k = n$ (i.e. $i_\pi(n, n) < n$) achieves strictly less than $\frac{1}{2}$: the contribution of the $k = n$ event drops from 2^{-n} to $2^{-n} \cdot v^\pi_{n-i_\pi(n,n)} < 2^{-n}$, while all other terms remain at most $\frac{1}{2} \cdot \binom{n}{k} 2^{-n}$, giving $v^\pi_n < \frac{1}{2}$.

3 Above the fair coin: $p > \frac{1}{2}$

Theorem 3.1. *For $p > \frac{1}{2}$ and all $n \geq 2$:*

(i) $w_{n,p} > w_{n-1,p}$, *i.e. $n \mapsto w_{n,p}$ is strictly increasing.*

(ii) $w_{n-1,p} < \dfrac{p^n}{p^n + q^n}$.

(iii) *Strategy* One *is optimal:* $w_{n,p} = a_{n,p}$.

Proof. We prove (i) and (ii) simultaneously by strong induction on n.

Base case $n = 2$. One computes $w_{1,p} = p$ and $w_{2,p} = p^2 + (1 - p^2 - q^2)p = 3p^2 - 2p^3$. Then $w_{2,p} - w_{1,p} = p(1 - p)(2p - 1) > 0$ since $p > \frac{1}{2}$. Also $w_{1,p} = p < p^2/(p^2 + q^2)$ iff $p^2 + q^2 > p$, which holds for $p \in (\frac{1}{2}, 1)$.

Inductive step. Assume (i) and (ii) hold for all indices up to $n - 1$. Since $n \mapsto w_{n,p}$ is increasing on $\{1, \ldots, n - 1\}$ by hypothesis, the suffix-maximum $\max_{j \leq m \leq n-1} w_{m,p} = w_{n-1,p}$ for every $j \in \{1, \ldots, n - 1\}$. Hence the Bellman equation (1) reduces to

$$w_{n,p} = p^n + (1 - p^n - q^n)\, w_{n-1,p},$$

which is exactly the recursion (2) for strategy One. This proves (iii).

For (i), $w_{n,p} > w_{n-1,p}$ iff $p^n(1 - w_{n-1,p}) > q^n w_{n-1,p}$, i.e. $w_{n-1,p} < p^n/(p^n + q^n)$, which is hypothesis (ii) at level $n - 1$.

For (ii) at level n, we need $w_{n,p} < p^{n+1}/(p^{n+1} + q^{n+1})$. Using the recursion and the inductive bound on $w_{n-1,p}$, a direct computation shows this reduces to the inequality $(q/p)^n < (q/p)^{n-1}$, which holds since $q/p < 1$. □

4 Below the fair coin: perturbation in $\delta = \frac{1}{2} - p$

For $p < \frac{1}{2}$ the structure of $n \mapsto w_{n,p}$ is far richer: numerical experiments show that the sequence develops local minima and maxima, and the optimal strategy is neither One nor All in general; see [7], Section 10.

To get analytical traction, we study the regime $p = \frac{1}{2} - \delta$ with $\delta > 0$ small.

4.1 The deficit and its recursion

Definition 4.1. For $p \in (0, 1)$ and $n \geq 1$, define the *deficit* $\Delta_{n,p} := \frac{1}{2} - w_{n,p}$.

By Theorem 2.1, $\Delta_{n,1/2} = 0$ for all n.

Proposition 4.2. *The deficit satisfies*

$$\Delta_{n,p} \;=\; \frac{q^n - p^n}{2} + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \min_{j \le m \le n-1} \Delta_{m,p}, \tag{5}$$

with $\Delta_{0,p} = -\frac{1}{2}$. *In particular:*

(i) *For* $p < \frac{1}{2}$: $\Delta_{n,p} > 0$ *for every* $n \ge 1$ *(i.e.* $w_{n,p} < \frac{1}{2}$*).*

(ii) *For* $p > \frac{1}{2}$: $\Delta_{n,p} < 0$ *for every* $n \ge 1$ *(i.e.* $w_{n,p} > \frac{1}{2}$*).*

Proof. Substitute $w_{m,p} = \frac{1}{2} - \Delta_{m,p}$ into the Bellman equation (1). The max on w_m becomes a min on Δ_m. The constant terms $\frac{1}{2} \cdot \big[\text{binomial weights}\big]$ cancel by the identity $p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \cdot \frac{1}{2} = \frac{1}{2} - \frac{1}{2}(q^n - p^n)$, leaving (5). The sign claims follow by induction using $q^n - p^n > 0$ (resp. < 0) and the positivity (resp. negativity) of the min terms. □

4.2 First-order expansion

Set $p = \frac{1}{2} - \delta$ with $\delta > 0$ small. Then $q = \frac{1}{2} + \delta$, and the constant term in (5) is

$$\frac{q^n - p^n}{2} = \frac{(\frac{1}{2} + \delta)^n - (\frac{1}{2} - \delta)^n}{2} = \frac{n\delta}{2^{n-1}} + O(\delta^3).$$

The binomial weights become $\binom{n}{j} 2^{-n} + O(\delta)$, and the min terms are $O(\delta)$ by Proposition 4.2. Thus $\Delta_{n,p} = O(\delta)$.

Proposition 4.3 (First-order coefficient)**.** *Write* $\Delta_{n,p} = c_n \delta + O(\delta^2)$ *as* $\delta \to 0^+$. *Then* $c_1 = 1$ *and for* $n \ge 2$*:*

$$c_n \;=\; \frac{n}{2^{n-1}} + \frac{1}{2^n} \sum_{j=1}^{n-1} \binom{n}{j} \min_{j \le m \le n-1} c_m. \tag{6}$$

Proof. Match first-order coefficients in (5). The constant term contributes $n/2^{n-1}$. Each binomial weight is $\binom{n}{j} 2^{-n} + O(\delta)$; the min of the $\Delta_m = c_m \delta + O(\delta^2)$ at first order is $(\min c_m)\delta + O(\delta^2)$. Products of $O(\delta)$ terms are $O(\delta^2)$. □

Example 4.4. The first few values of c_n:

$$c_1 = 1, \quad c_2 = \tfrac{3}{2}, \quad c_3 = \tfrac{27}{16}, \quad c_4 = \tfrac{111}{64}, \quad c_5 = \tfrac{3555}{2048} \approx 1.7358, \quad c_6 = \tfrac{113337}{65536} \approx 1.7293.$$

The sequence increases on $\{1, \ldots, 5\}$ and decreases for $n \ge 6$. In particular $c_4 = \frac{111}{64} > c_6$ and $c_5 > c_6$, both of which we use below as base cases.

4.3 Collapse, lower bound, and linear recursion

The four results below — the asymptotic collapse of the min in (6) (Lemma 4.5), the uniform lower bound $c_m \geq \frac{27}{16}$ (Lemma 4.6), the strict decrease of $(c_n)_{n\geq 5}$ (Lemma 4.7), and the linear recursion for $n \geq 7$ (Proposition 4.8) — are mutually entangled: each, taken alone, depends on parts of the others. Stating them in isolation produces an apparent circular dependency. We therefore state all four results first and then prove them simultaneously by induction on n, with the IH at level $< n$ supplying exactly what is needed at level n.

Lemma 4.5 (Asymptotic collapse)**.** *For all $n \geq 7$ and $j \in \{1, \ldots, n-1\}$:*

$$\min_{j\leq m\leq n-1} c_m \;=\; \begin{cases} c_j & \text{if } j \in \{1,2,3\}, \\ c_{n-1} & \text{if } j \in \{4,\ldots,n-1\}. \end{cases} \tag{7}$$

Lemma 4.6. *$c_m \geq \frac{27}{16}$ for every $m \geq 4$.*

Lemma 4.7. *The sequence $(c_n)_{n\geq 5}$ is strictly decreasing.*

Proposition 4.8 (Linear recursion)**.** *For $n \geq 7$:*

$$c_n \;=\; A_n + (1 - B_n)\, c_{n-1}, \tag{8}$$

where

$$A_n := \frac{n}{2^{n-1}} + \frac{n\,c_1 + \binom{n}{2}c_2 + \binom{n}{3}c_3}{2^n}, \qquad B_n := \frac{2 + n + \binom{n}{2} + \binom{n}{3}}{2^n}. \tag{9}$$

Moreover the algebraic identity

$$A_n - \tfrac{27}{16}B_n \;=\; -\frac{3(n^2 - 15n + 36)}{32 \cdot 2^n} \tag{10}$$

holds for every $n \geq 1$ (independently of the linear recursion).

Joint proof of Lemmas 4.5, 4.6, 4.7 and Proposition 4.8. The identity (10) is a direct algebraic computation from (9) using $\binom{n}{2} = \frac{n(n-1)}{2}$ and $\binom{n}{3} = \frac{n(n-1)(n-2)}{6}$; we use it freely below.

For $n \geq 4$ let $P(n)$ denote the conjunction of the following four claims (each understood vacuously when its index range is empty):

(a) Lemma 4.5 holds at level n (when $n \geq 7$);

(b) $c_n \geq \frac{27}{16}$;

(c) $c_n < c_{n-1}$ (when $n \geq 6$);

(d) the linear recursion (8) holds at level n (when $n \geq 7$).

We prove $P(n)$ for all $n \geq 4$ by strong induction on n.

Base cases $n \in \{4, 5, 6\}$. By Example 4.4, $c_4 = \frac{111}{64}$, $c_5 = \frac{3555}{2048}$, $c_6 = \frac{113337}{65536}$, all of which exceed $\frac{27}{16} = \frac{108}{64}$, so (b) holds. Also $c_5 > c_6$, so (c) holds at $n = 6$. Claims (a) and (d) are vacuous for $n \leq 6$.

Inductive step. Fix $n \geq 7$ and assume $P(m)$ for all $4 \leq m < n$. In particular: by IH(b), $c_m \geq \frac{27}{16}$ for $4 \leq m \leq n-1$; by IH(c), $c_5 > c_6 > \cdots > c_{n-1}$. We now establish (a)–(d) at level n.

(a) Collapse. For $j \in \{1,2,3\}$: from $c_1 = 1 < c_2 = \frac{3}{2} < c_3 = \frac{27}{16}$ (Example 4.4) and $c_m \geq \frac{27}{16} = c_3$ for $m \geq 4$ (IH(b)), the minimum over $\{j, \ldots, n-1\}$ is c_j.

For $j \in \{4, \ldots, n-1\}$: by IH(c), $(c_m)_{m\geq 5}$ is strictly decreasing, so $c_5 > c_6 > \cdots > c_{n-1}$, and in particular $c_m \geq c_{n-1}$ for $m \in \{5, \ldots, n-1\}$. For $m = 4$, $c_4 = \frac{111}{64} > c_6$ (Example 4.4) and $c_6 \geq c_{n-1}$ (IH(c)), so $c_4 \geq c_{n-1}$ as well. Hence the minimum over $\{j, \ldots, n-1\}$ for $j \geq 4$ is c_{n-1}.

(d) Linear recursion. Substituting the collapse (a) just proved into (6) and splitting the sum at $j = 3$ yields (8)–(9).

(b) Lower bound. From (d) and (10),

$$c_n - \tfrac{27}{16} \;=\; \big(A_n - \tfrac{27}{16}B_n\big) + (1 - B_n)\big(c_{n-1} - \tfrac{27}{16}\big).$$

By IH(b), $c_{n-1} - \frac{27}{16} \geq 0$. Note also $0 < B_n < 1$ for all $n \geq 7$: the numerator $2 + n + \binom{n}{2} + \binom{n}{3}$ in (9) is strictly less than $2^n = \sum_{j=0}^{n} \binom{n}{j}$, the difference being $\sum_{j=4}^{n-1} \binom{n}{j} > 0$. For $n \in \{7, \ldots, 12\}$, the quadratic $n^2 - 15n + 36$ is non-positive on $\{3, \ldots, 12\}$ (its roots are 3 and 12), so $A_n - \frac{27}{16}B_n \geq 0$ by (10), and (b) follows. For $n \geq 13$ we argue cumulatively: define $\varepsilon_m := c_m - \frac{27}{16}$ and $\delta_k := 3(k^2 - 15k + 36)/(32 \cdot 2^k)$, so (d) gives $\varepsilon_m = (1 - B_m)\varepsilon_{m-1} - \delta_m$. The starting value $\varepsilon_{12} > \frac{1}{60}$ is verified by exact rational computation; the cumulative negative contribution is bounded by $\sum_{k\geq 13} \delta_k \leq \frac{297}{65536} < \frac{1}{200}$ (using $\sum_{k\geq 13} k^2/2^k = \frac{99}{2048}$); meanwhile $\prod_{m\geq 13}(1 - B_m) > \frac{7}{8}$ (using $\sum_{m\geq 13} B_m < \frac{1}{8}$). So $\varepsilon_n > \frac{1}{60} \cdot \frac{7}{8} - \frac{1}{200} = \frac{23}{2400} > 0$ for $n \geq 13$.

(*Note for an inductive treatment:* the sequence $(\delta_k)_{k\geq 13}$ is not monotone — one computes $\delta_{14}/\delta_{13} = \frac{11}{10}$ — but its ratio decreases through 1 at $k = 14$ and tends to $\frac{1}{2}$. An iterated proof of the bound on $\sum_{k\geq 13} \delta_k$ should therefore split at $k = 14$; the closed-form route above sidesteps this subtlety.)

(c) Strict decrease. By (d), $c_n - c_{n-1} = A_n - B_n c_{n-1}$. Using $c_{n-1} \geq \frac{27}{16}$ (IH(b)) and $B_n > 0$, this is $\leq A_n - \frac{27}{16}B_n < 0$ for $n \geq 13$ by (10) ($n^2 - 15n + 36 > 0$ there). For $n \in \{7, \ldots, 12\}$, $c_n < c_{n-1}$ is verified by direct rational computation. This completes the inductive step. □

4.4 The limit

Theorem 4.9 (Existence and value of the limit)**.** *The limit $L := \lim_{n\to\infty} c_n$ exists, and for any $n_0 \geq 7$:*

$$L \;=\; c_{n_0-1} \prod_{m\geq n_0} (1 - B_m) \;+\; \sum_{k\geq n_0} A_k \prod_{m>k} (1 - B_m). \tag{11}$$

Both the infinite product and the series converge at geometric rate since $A_n, B_n = O(n^3/2^n)$. Numerically,

$$L \;=\; 1.70347176087173673645\ldots \tag{12}$$

Proof. Existence. By Lemmas 4.6 and 4.7, $(c_n)_{n\geq 5}$ is strictly decreasing and bounded below by $\frac{27}{16}$, hence $L := \lim_{n\to\infty} c_n$ exists.

Explicit form. Fix $n_0 \geq 7$. Iterating the linear recursion (8) from n_0 up to n yields, for every $n \geq n_0$,

$$c_n \;=\; c_{n_0-1} \prod_{m=n_0}^{n} (1 - B_m) \;+\; \sum_{k=n_0}^{n} A_k \prod_{m=k+1}^{n} (1 - B_m). \tag{13}$$

We pass to the limit $n \to \infty$ in each summand.

Step 1 (infinite product). Since $B_m = O(m^3/2^m)$, the series $\sum_{m \geq n_0} B_m$ converges absolutely. Because $0 < B_m < 1$ for all $m \geq 7$ (shown in the proof of Lemma 4.6), the partial products $\prod_{m=n_0}^{n}(1 - B_m)$ converge as $n \to \infty$ to a strictly positive limit $\prod_{m \geq n_0}(1 - B_m)$; cf. Knopp [4, §29].

Step 2 (series of products). For each fixed $k \geq n_0$ the same argument gives $\prod_{m=k+1}^{n}(1 - B_m) \to \prod_{m>k}(1 - B_m)$ as $n \to \infty$. Moreover every factor lies in $(0, 1)$, so each finite product lies in $(0, 1]$, and

$$\left| A_k \prod_{m=k+1}^{n} (1 - B_m) \right| \;\leq\; |A_k| \qquad (n \geq k). \tag{14}$$

Since $A_k = O(k^3/2^k)$, the series $\sum_{k \geq n_0} |A_k|$ converges. The bound (14) therefore supplies a summable dominant, so by Tannery's theorem — the dominated convergence theorem for series, see Knopp [4, §41] — limit and summation commute:

$$\lim_{n \to \infty} \sum_{k=n_0}^{n} A_k \prod_{m=k+1}^{n} (1 - B_m) \;=\; \sum_{k \geq n_0} A_k \prod_{m>k} (1 - B_m).$$

Conclusion. Taking $n \to \infty$ in (13) and using Steps 1 and 2 gives (11). Both the product and the series converge at geometric rate since $A_n, B_n = O(n^3/2^n)$. □

4.5 Shape of the optimal-value sequence near $p = \frac{1}{2}$

Corollary 4.10. *For $\delta > 0$ sufficiently small and $p = \frac{1}{2} - \delta$:*

(i) $w_{n-1,p} - w_{n,p} = (c_n - c_{n-1})\delta + O(\delta^2)$.

(ii) *The sequence $n \mapsto w_{n,p}$ has, to first order in δ, a strict local minimum at $n = 5$ (since $c_5 > c_4$ and $c_5 > c_6$).*

(iii) *There is no local maximum at first order: c_n is eventually decreasing, so $w_{n,p}$ is eventually increasing.*

Remark 4.11. Numerical computation shows that the sequence $n \mapsto w_{n,p}$ does develop strict local maxima for p bounded away from $\frac{1}{2}$ (e.g. at $n = 9$ for $p = 0.42$, verified by exact rational arithmetic). This is consistent with the corollary: local maxima are a *non-perturbative* phenomenon invisible to the first-order expansion. Van Doorn [7], Section 10, conjectures that the optimal strategy for $p < \frac{1}{2}$ is "infinitely complex."

5 Numerical experiments

The first-order analysis of Section 4 describes the shape of $n \mapsto w_{n,p}$ only in a shrinking neighbourhood of $p = \frac{1}{2}$. Corollary 4.10 guarantees, for every sufficiently small $\delta > 0$, a unique local minimum at $n = 5$ and no local maximum. Numerical computation of $w_{n,p}$ confirms this picture for p close to $\frac{1}{2}$, and shows that local maxima emerge for p bounded away from $\frac{1}{2}$.

All values of $w_{n,p}$ below are computed by the exact Bellman recursion (1), evaluated in arbitrary-precision rational (or 80-decimal-digit) arithmetic via `mpmath`. The code is available in the `simulation/` subdirectory of the repository.

5.1 Optimal winning probability for $p < \frac{1}{2}$

Figure 1 plots $w_{n,p}$ against n for five values of p. For $p = 0.49$ (first-order regime) the sequence bottoms out at $n = 5$ and increases thereafter, in agreement with Corollary 4.10. For $p = 0.42$ and $p = 0.45$ a non-monotone structure emerges: the sequence first decreases, passes through a local minimum, then rises to a local *maximum* before decreasing again and converging.

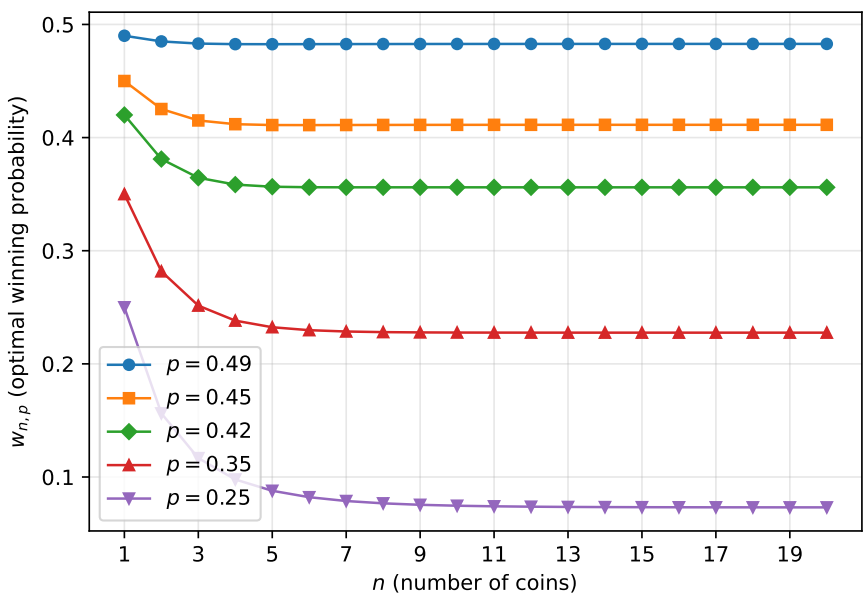

Figure 1: The optimal winning probability $w_{n,p}$ as a function of n for five values of $p < \frac{1}{2}$. Computed by the exact Bellman recursion in 80-digit arithmetic. For $p = 0.49$ only the predicted local minimum at $n = 5$ is visible; for $p = 0.42$ and $p = 0.45$ local maxima appear. For $p = 0.35$ and $p = 0.25$ no local extrema occur within $n \leq 20$.

The non-monotone behaviour is invisible in the plot at the scale of Figure 1, because the oscillations are small compared with the total variation of $w_{n,p}$ in n. Figure 2 zooms in on $p = 0.42$ and $p = 0.45$ and highlights the local minima and maxima found by direct inspection of the sequence.

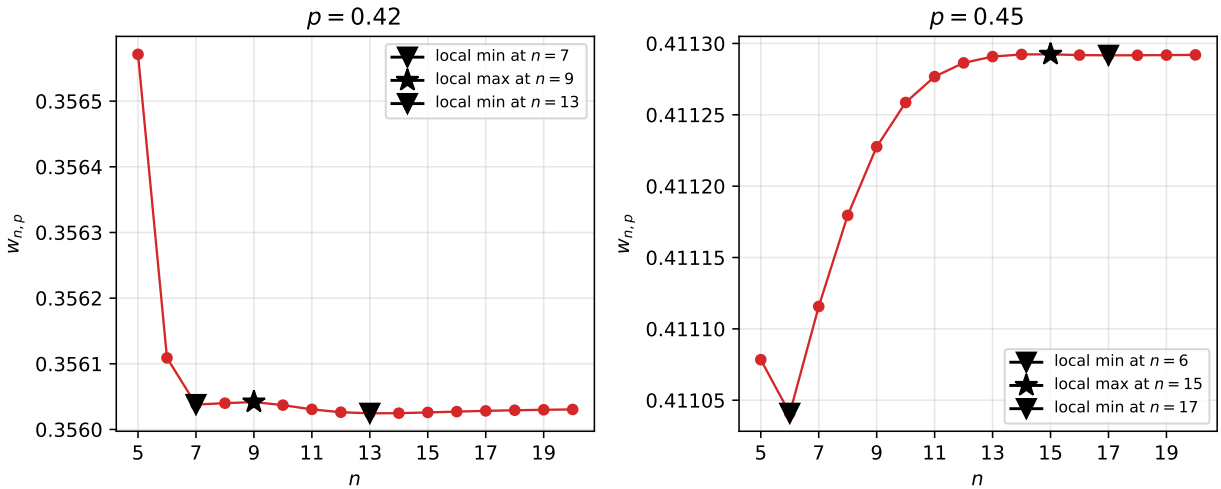

Figure 2: Zoomed-in view of $w_{n,p}$ for $p \in \{0.42, 0.45\}$ and $n \in [5, 20]$. Stars mark local maxima, inverted triangles mark local minima. At $p = 0.42$ there is a local maximum at $n = 9$; at $p = 0.45$ there is a local maximum at $n = 15$.

Table 1 lists the numerical values of $w_{n,p}$ for $1 \leq n \leq 20$ and the five values of p considered in Figure 1. The local extrema are tabulated in Table 2.

5.2 Discussion

Two features of the numerical data are worth flagging.

(a) Local maxima are a non-perturbative phenomenon. Corollary 4.10(iii) rules out local maxima of $n \mapsto w_{n,p}$ *to first order in* $\delta = \frac{1}{2} - p$: for any fixed n, there is a $\delta_0(n) > 0$ below which the predicted sign of $c_{n+1} - c_n < 0$ forces $w_{n,p} < w_{n+1,p}$. The local maximum at $n = 9$ for $p = 0.42$ ($\delta = 0.08$), and the one at $n = 15$ for $p = 0.45$ ($\delta = 0.05$), correspond to the threshold $\delta_0(n)$ being *larger* than the actual perturbation size at that n, so higher-order terms dominate. Numerically, $\delta_0(n)$ appears to decay rapidly in n (we see no local maxima at all for $p \leq 0.35$ within $n \leq 20$), which is consistent with the expected geometric decay of the first-order constants A_n and $B_n = O(n^3/2^n)$.

(b) The first-order picture holds robustly near $p = \frac{1}{2}$. For $p = 0.49$ ($\delta = 0.01$), the only local extremum within $n \leq 20$ is the predicted minimum at $n = 5$, with no spurious oscillations. This confirms numerically that the first-order range-of-validity extends at least to $\delta = 0.01$. Pinning down the size of $\delta_0(n)$ as a function of n remains open; we conjecture that it decays roughly like C/n^2 for some absolute constant C.

Remark 5.1. The picture for $p \leq 0.35$ in our data — a strictly decreasing sequence with no local extrema within $n \leq 20$ — is consistent with, but does not prove, monotone decrease on all of $\mathbb{N}$. Van Doorn [7] proves that strategy ALL is *not* optimal for $p < \frac{1}{2}$ (there exists n with $w_{n,p} \neq b_{n,p}$); equivalently, the sequence $n \mapsto w_{n,p}$ is not monotone non-increasing. The failure of monotonicity is, however, not always detectable as a local extremum in n; it can manifest as strict inequality $w_{n,p} > b_{n,p}$ without the sequence $w_{n,p}$ itself having oscillations. Investigating this gap is beyond the scope of the present paper.

n	$p = 0.49$	$p = 0.45$	$p = 0.42$	$p = 0.35$	$p = 0.25$
1	0.49000000	0.45000000	0.42000000	0.35000000	0.25000000
2	0.48500200	0.42525000	0.38102400	0.28175000	0.15625000
3	0.48309103	0.41514272	0.36450348	0.25147259	0.11669922
4	0.48258194	0.41184019	0.35845021	0.23820137	0.09781647
5	**0.48254059**	0.41107840	0.35657129	0.23236999	0.08782906
6	0.48259223	**0.41104082**	0.35610901	0.22977285	0.08215652
7	0.48264193	0.41111568	**0.35603793**	0.22859504	0.07878468
8	0.48268806	0.41117954	0.35604004	0.22805232	0.07672145
9	0.48272442	0.41122762	*0.35604176*	0.22779987	0.07543431
10	0.48275111	0.41125863	0.35603705	0.22768230	0.07462024
11	0.48276987	0.41127677	0.35603062	0.22762785	0.07410012
12	0.48278262	0.41128638	0.35602626	0.22760288	0.07376520
13	0.48279104	0.41129079	**0.35602460**	0.22759155	0.07354821
14	0.48279647	0.41129231	0.35602484	0.22758643	0.07340694
15	0.48279990	*0.41129237*	0.35602596	0.22758409	0.07331459
16	0.48280201	0.41129184	0.35602720	0.22758299	0.07325404
17	0.48280329	**0.41129168**	0.35602833	0.22758245	0.07321422
18	0.48280406	0.41129173	0.35602926	0.22758218	0.07318798
19	0.48280451	0.41129185	0.35602997	0.22758204	0.07317065
20	0.48280478	0.41129198	0.35603049	0.22758197	0.07315919

Table 1: Numerical values of $w_{n,p}$ for $n \in [1, 20]$ and five values of $p < \frac{1}{2}$. Local minima are marked in **bold**, local maxima in *italics*.

A Formal verification in Lean 4

All results of this paper have been formally verified in Lean 4 using the Mathlib library. The formalization is publicly available at

`https://github.com/pfaffelh/coins`

and builds successfully against Mathlib with no open goals. The proofs use only the three standard Lean foundational axioms (`propext`, `Classical.choice`, `Quot.sound`); no additional axioms, `sorry`s, `native_decide`, `unsafe` blocks, or `opaque` definitions appear.

Table 3 lists every numbered theorem, proposition, lemma, and corollary stated in the main text together with its corresponding Lean theorem. Each hyperlink points to the exact line of the current pinned revision (commit `b98f5ba`).

The dominated-convergence step in the proof of Theorem 4.9 (Step 2) is discharged in Lean by Mathlib's `tendsto_tsum_of_dominated_convergence` (Tannery's theorem), and the infinite-product convergence in Step 1 uses `Real.multipliable_one_add_of_summable`. These are exactly the two analytic ingredients cited in the proof of Theorem 4.9.

p	Local minima at $n =$	Local maxima at $n =$
0.49	5	—
0.45	6, 17	15
0.42	7, 13	9
0.35	—	—
0.25	—	—

Table 2: Local extrema of $n \mapsto w_{n,p}$ for $n \in [2, 19]$.

Manuscript	**Lean theorem**	**Location**
Theorem 2.1	`w_half`	Optimal.lean:102
Theorem 3.1 (joint)	`above_half`	Above.lean:108
Proposition 4.2	`deficit_succ`	Perturbation.lean:301
Proposition 4.3	`deficit_first_order`	Perturbation.lean:2503
Lemma 4.5 (low)	`suffMin_collapse_low`	Perturbation.lean:1501
Lemma 4.5 (high)	`suffMin_collapse_high`	Perturbation.lean:1507
Lemma 4.6	`c_ge_27_16_full`	Perturbation.lean:1483
Lemma 4.7	`c_strict_anti_from_five`	Perturbation.lean:1488
Proposition 4.8	`c_linear_rec`	Perturbation.lean:1495
Theorem 4.9 (existence)	`c_limit_exists`	Perturbation.lean:1517
Theorem 4.9 (formula)	`c_limit_formula`	Perturbation.lean:2065
Corollary 4.10 (i)	`w_gap_first_order`	Perturbation.lean:2811
Corollary 4.10 (ii)	`w_local_min_at_five`	Perturbation.lean:2847
Corollary 4.10 (iii)	`no_first_order_local_max`	Perturbation.lean:2948

Table 3: Correspondence between manuscript results and their Lean formalization. Links point to the exact line of commit `b98f5ba` on the repository's `main` branch.

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